

3 Two progressive water waves X and Y travel along a straight line from point A to point B. The variation of displacement of the waves with distance from A at an instant in time is shown in Fig. 3.1.

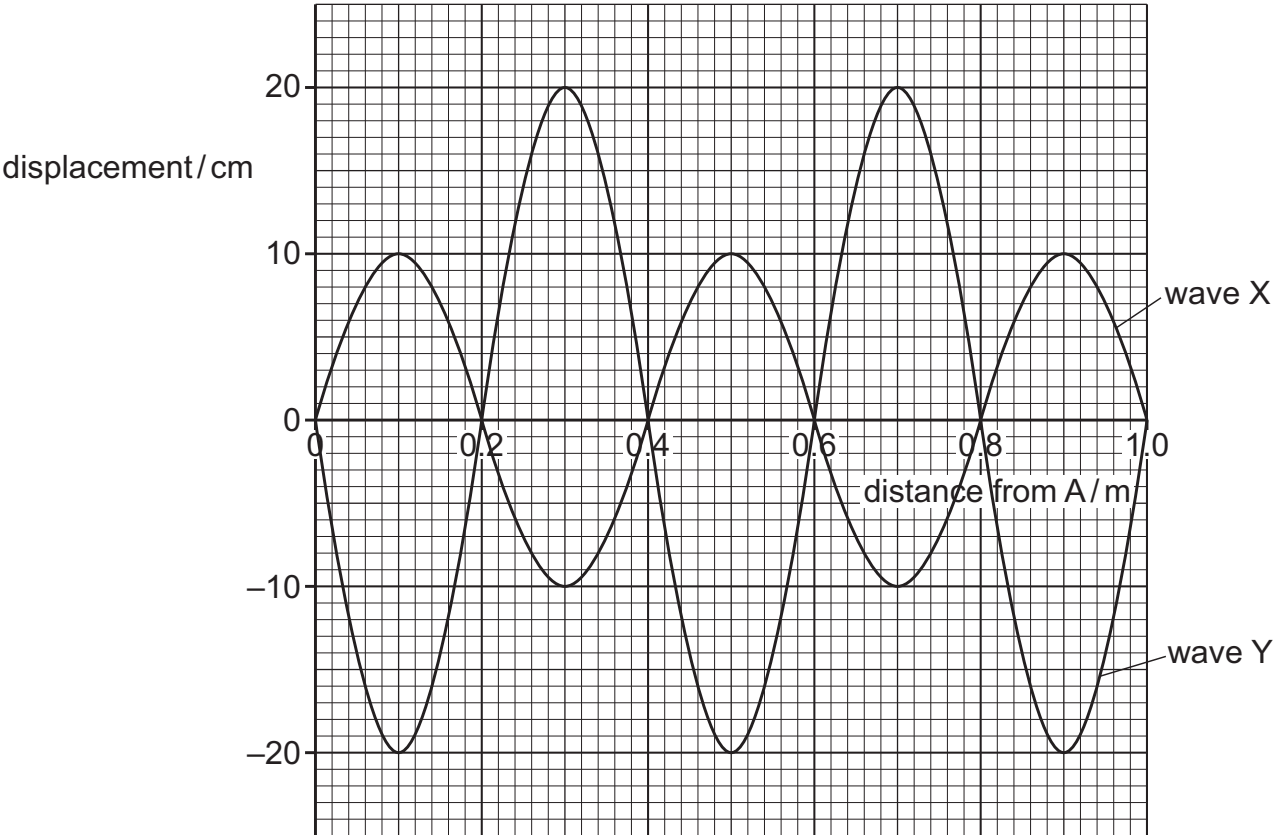


Fig. 3.1

(a) State the amplitude of wave X.

amplitude = cm [1]

(b) Both waves have frequency 16 Hz.

(i) Determine the speed of wave X.

speed = ms⁻¹ [2]

(ii) State and explain whether X and Y are coherent.

.....

 [1]

(c) Wave X and wave Y superpose to form a resultant wave.

On Fig. 3.2, sketch the variation of displacement of the resultant wave with distance from A at the instant of time shown in Fig. 3.1.

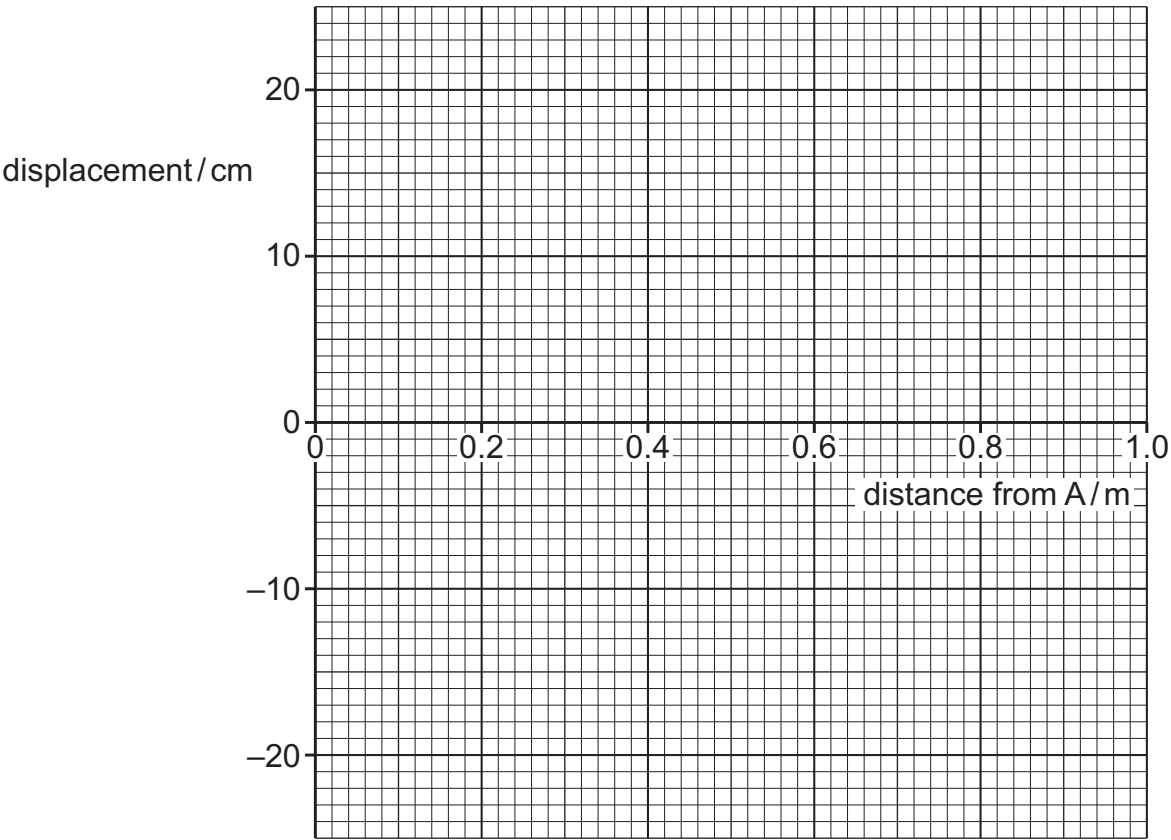


Fig. 3.2

[2]

(d) The intensity of wave X is I_X . The intensity of wave Y is I_Y .

Use Fig. 3.1 to determine the ratio $\frac{I_X}{I_Y}$.

ratio = [2]